TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Mechanical Equivalent of Heat by the Electrical Calorimeter

Joule's exchange rate between electrical work and heat: pour a measured number of joules into water and count the calories that appear — their ratio is J.

PHY-1204 · PHYSICS-II SESSIONAL · EXPERIMENT 7 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to determine J, the mechanical equivalent of heat (joules per calorie), by electrical heating. A current through a coil immersed in the calorimeter's water converts electrical energy VIt (joules) entirely into heat. That heat is absorbed by the water (mass m, specific heat 1 cal g^{-1} $°C^{-1}$) and by the calorimeter vessel and fittings, whose appetite for heat is captured by their **water equivalent** W (the mass of water that would absorb the same heat per degree). If the temperature rises by Δθ, the heat that appeared is

$$H = (m + W)\,\Delta\theta \quad \text{calories}$$

and the equivalent follows from energy conservation, VIt joules ≡ H calories:

$$J = V\,I\,t \,/\, ((m + W)\,\Delta\theta) \quad \text{J/cal}$$

The accepted value is 4.18 J/cal. The experiment's enemies are heat losses to the room during the run — they make less heat appear in the water than the electricity supplied, so J comes out high — countered by insulation, a modest temperature rise, and a cooling correction estimated from the temperature fall after switch-off.

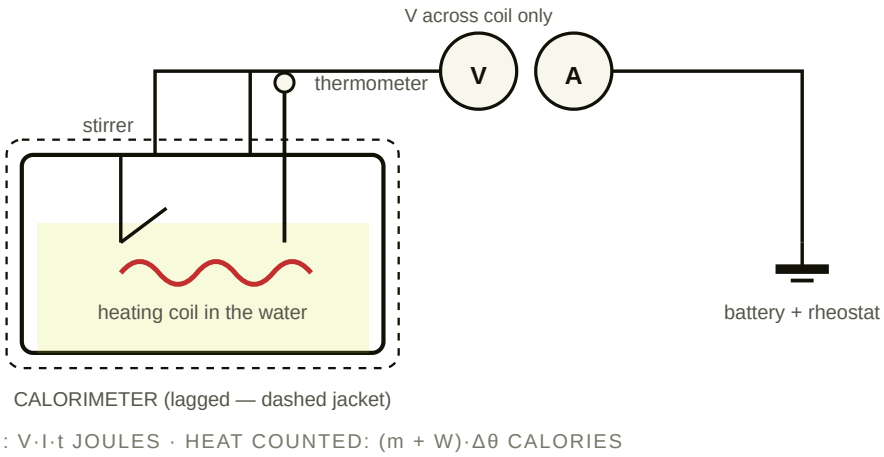


Figure 1 — The electrical calorimeter. The voltmeter stands across the coil alone (so lead drops are not counted), the ammeter in series reads the coil current, and lagging plus stirring fight the two great errors: lost heat and uneven temperature.

2. Apparatus and procedure

- **Apparatus:** calorimeter with heating coil and lagging, thermometer (0.1 °C), stirrer, voltmeter, ammeter, battery/DC supply, rheostat, key, stopwatch, balance.
- Weigh the empty calorimeter with stirrer; add a known mass of water (about 200 g, enough to cover the coil fully), weigh again to get m. Take the water equivalent W of the vessel from its mass and material (or a stated value).
- Connect the coil in series with ammeter, rheostat and supply; voltmeter directly across the coil terminals. Stir and record the initial temperature θ_1 (water a few degrees below room temperature is a good start — early gains from the room offset later losses).
- Close the key and start the watch together. Hold V and I steady (trim the rheostat), stirring gently throughout. After about 10 minutes, open the key, note t, and keep stirring while recording the highest temperature reached, θ_2.
- Cooling correction: watch the temperature fall for a few minutes after switch-off; the fall rate near θ_2 estimates the loss during the run and the correction to add to $\Delta\theta = \theta_2 - \theta_1$.
- Compute J with and without the correction.

3. Worked calculation

Run: V = 6.0 V, I = 2.0 A, t = 600 s → energy in = 7,200 J. Water m = 200 g; water equivalent W = 20 g; rise measured $\Delta\theta$ = 7.5 °C.

$$J = 7200 / (220 \times 7.5) = 4.36 \text{ J/cal (uncorrected)}$$

The temperature fell 0.4 °C-worth during the run's timescale, so the corrected rise is 7.9 °C:

$$J = 7200 / (220 \times 7.9) = 4.14 \text{ J/cal (corrected)} \text{ — against the standard } 4.18$$

4. Precautions

- The coil must stay fully immersed and must never be switched on in air — it would overheat and its resistance (hence the true V and I) would drift.
- Stir continuously but gently: the thermometer must read the water's true mean temperature, and violent stirring adds its own (tiny) work and splashes.
- Voltmeter across the coil only; ammeter and rheostat drops are not heat in the water and must not be paid for.
- Start below room temperature and finish about as far above it, so room gains and losses roughly cancel; add the cooling correction regardless.

Practice (viva preparation)

1. What is the water equivalent of a calorimeter, and how does it enter the calculation?
2. A 12 V, 3 A coil runs for 5 minutes in 300 g of water with W = 25 g. Predict the temperature rise (take J = 4.18).
3. Why does neglecting heat loss make J come out too *large*, not too small?
4. The voltmeter is mistakenly connected across coil + ammeter. What error follows?
5. Why begin with the water a few degrees below room temperature?

Answer key

1. The water equivalent W is the mass of water that absorbs the same heat per degree as the calorimeter, stirrer and immersed fittings. The heat that "appears" is therefore $(m + W)\Delta\theta$ calories — the vessel drinks alongside the water and must be counted.
2. Energy = $12 \times 3 \times 300 = 10{,}800$ J = 10,800/4.18 = 2,584 cal. $\Delta\theta = 2584/(325) \approx 7.9$ °C.
3. J is computed as (electrical energy in)/(heat counted in the water). Lost heat never registers in $\Delta\theta$, so the denominator is too small while the numerator is exact — the quotient is inflated.
4. The measured V then includes the ammeter's voltage drop, so VIt overstates the energy delivered to the water; J is inflated again — the voltmeter must hug the coil's own terminals.
5. Below room temperature the calorimeter gains heat from the room; above it, it loses heat. A symmetric run — starting as far below as it ends above — makes the gain and loss nearly cancel over the run.